

# Noise Prediction along Bridges using CNOSSOS-EU and ASJ-RTN Models

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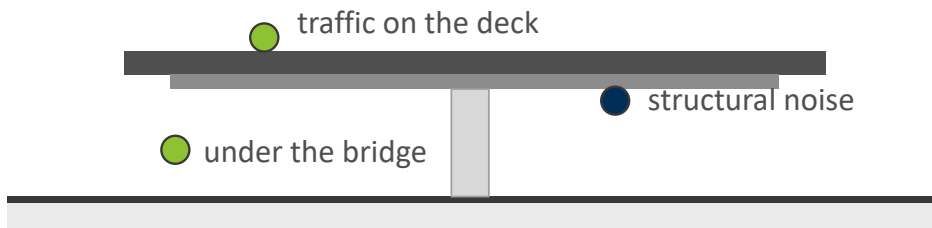
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Yokohama Bay Bridge ©Metropolitan Expressway Company Limited.

# Why bridges

- **Elevated roads sit right next to people**
  - homes beside the deck
- **Prediction needs three sources × three ingredients**
  - sources: deck, structure, under the bridge
  - ingredients: position, level, propagation → nine cells
- **Few cell is fully covered today**
  - CNOSSOS-EU: level of the traffic source
  - its position and propagation: conditional only
  - structural noise: no model at all



|                                   | Source position | Source level | Propagation |
|-----------------------------------|-----------------|--------------|-------------|
| Source on the deck                | △               | ✓            | △           |
| Source under / beside the deck    | △               | ✓            | △           |
| Structural noise (deck underside) | X               | X            | X           |

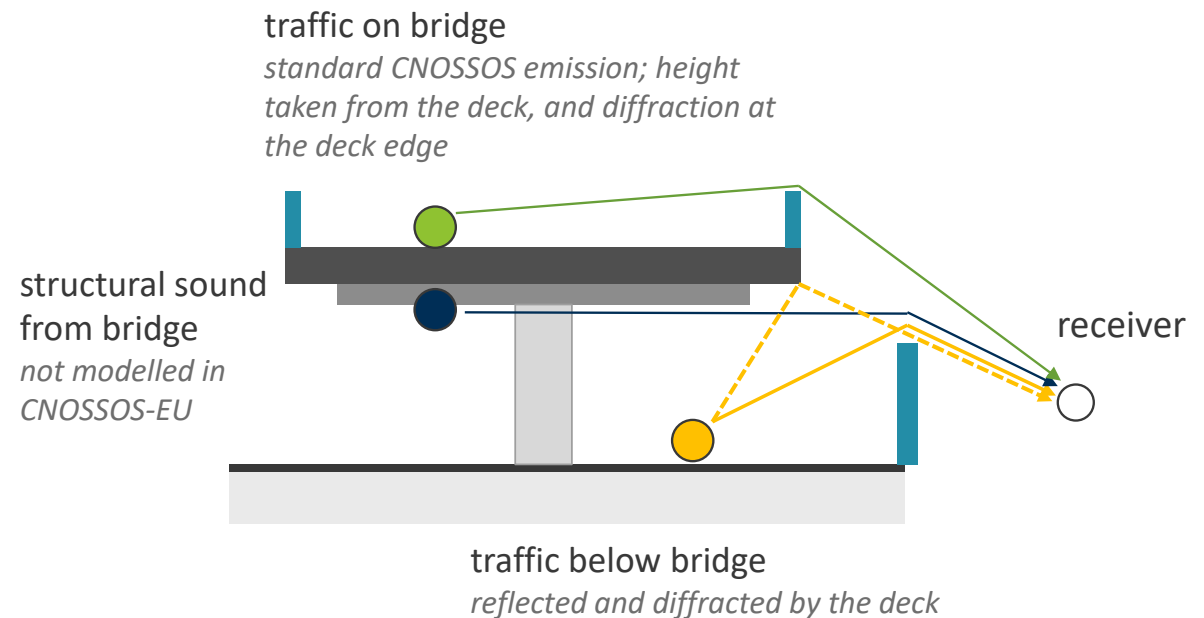
✓ covered by CNOSSOS-EU

△ conditional — height from terrain; deck as a plain screen

X not covered — and not implemented in NoiseModelling

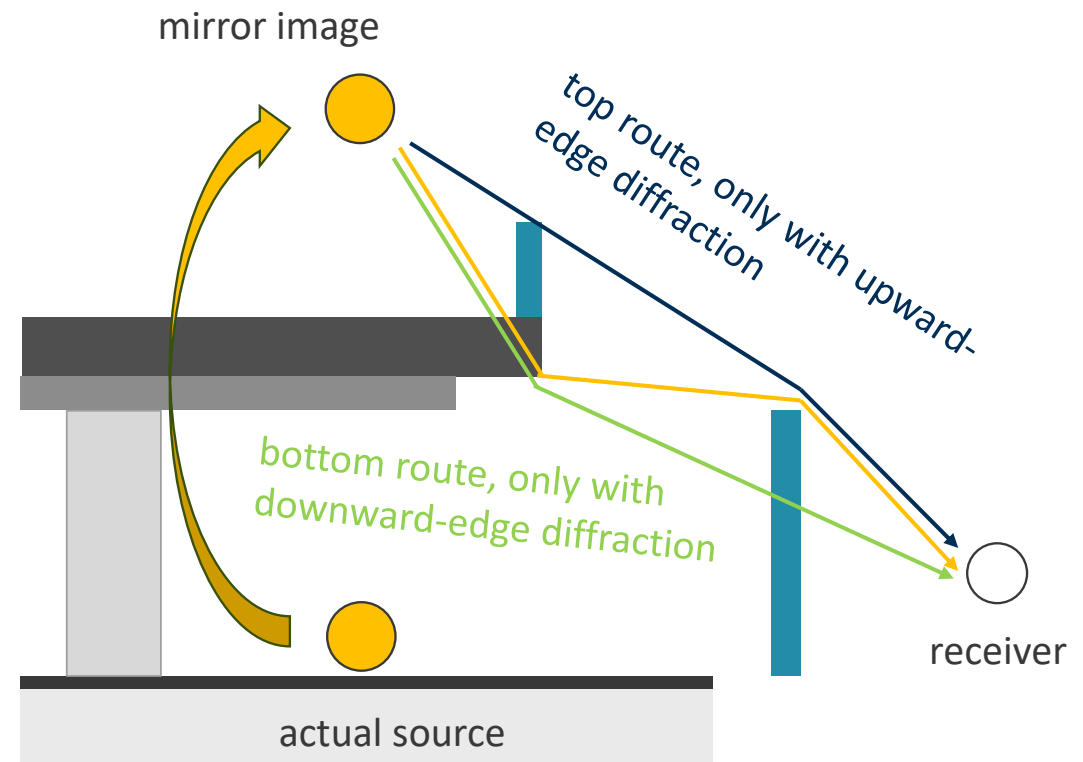
# Sound sources related to a bridge deck

- **Three sources, three different characters**
  - see the vertical section on the right
- **Deck object for the noise prediction**
  - ground reference for the source standing on it
  - obstacle for propagation:
    - screen with upward edge
    - (reflector and screen with downward edge)
- **Defining the deck widens the covered range**
  - positions of the on-deck source and the structural source become complete
  - still open: the propagation path from source under the deck and the structural emission level



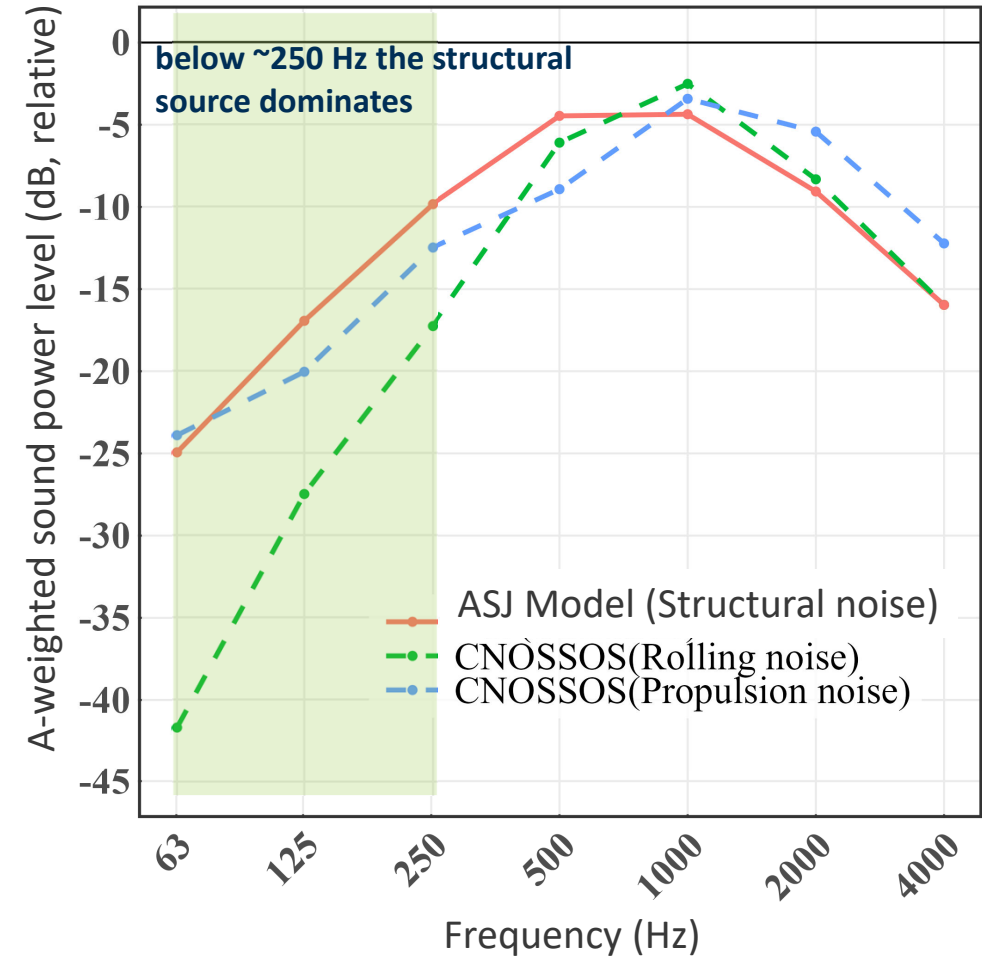
# A source under the bridge: reflection and (double) diffraction

- **A ray only diffracts over an obstacle at CNOSSOS-EU**
  - paths are built from the upper convex hull; a bridge deck is the first obstacle a ray can also pass under
- **Mirror image and downward-edge diffraction**
  - reflection: a mirror image of the source across it
  - diffraction: downward-edge at the bridge edge
  - passing the deck's lower edge can put a further obstacle in the path: double diffraction on one ray
- **Split into a bottom route and a top route**
  - a `CnossosPath` now holds two child `CnossosPath`: bottom route (downward edge) and top route (upward edge)
  - two ordinary single diffractions, combined the way the ASJ RTN-Model treats multiple diffraction



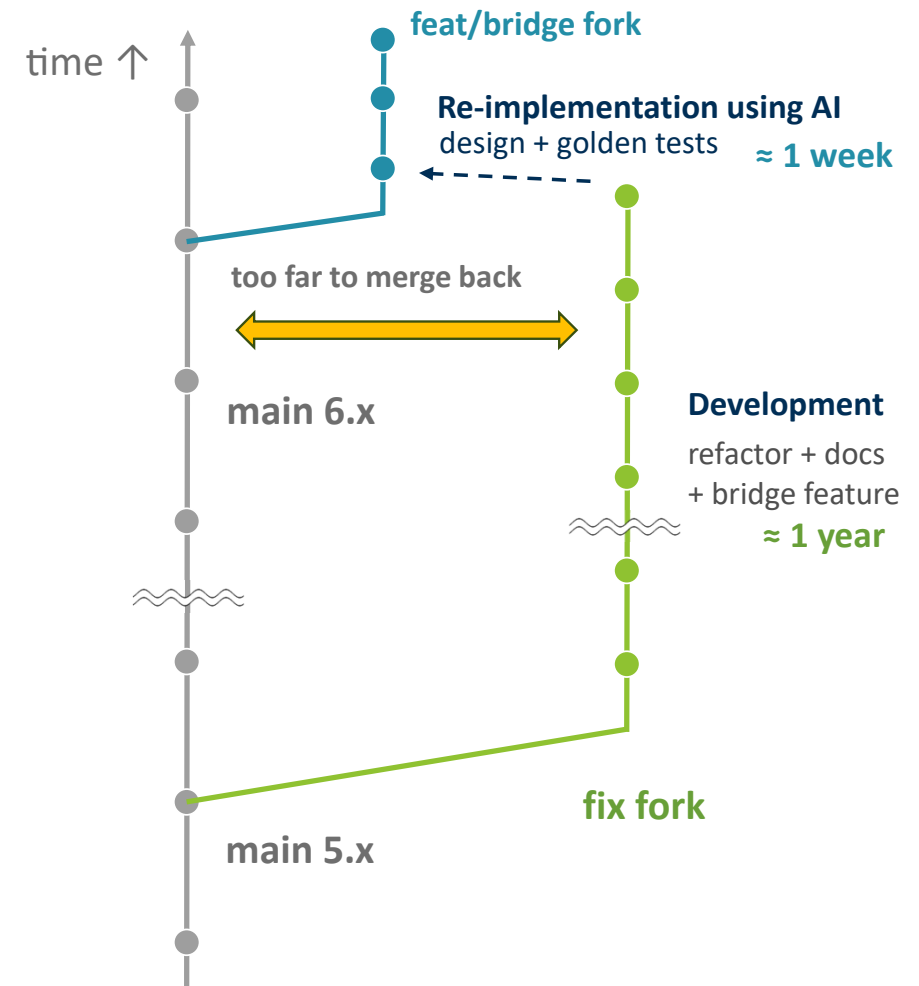
# Structural noise: another source on the deck underside

- **The missing cell: the structural emission level**
  - the deck and girders radiate noise themselves, vibrating under traffic: outside the scope of CNOSSOS-EU
- **ASJ RTN-Model 2023 provides that emission**
  - number of heavy- and middle-weight vehicles and traffic speed
  - 7 girder / slab construction types (steel, concrete, ...)
- **Added structural noise just under the bridge deck**
  - using traffic noise geometry and attributes



# How this was built

- **fix fork: from NoiseModelling main 5.x**
  - about one year of hands-on development: refactoring, developer docs, bridge feature
- **The fork was too far to merge back to main 6.x**
  - the design was proven, but the diff was no longer reusable
- **Re-implemented on current main with an AI coding agent**
  - the physics and the design were already settled; reproducing them against a moving upstream was large but mechanical work
  - one concentrated pass of about a week with Claude, using the fork as the design and golden-test reference



# Status: ready to propose

Three stacked, incrementally reviewable PRs on current main  
— all implemented, tested, with draft PR bodies written

## v0

### Bridge object

The deck becomes ground reference and screens sound from above.

**fills: position of the on-deck and under-deck sources; propagation of the on-deck source**



## v1

### Sound ray under the deck

Deck-underside reflection and bottom/top diffraction are defined.

**fills: propagation of the under-deck source**



## v2

### Structural Sources below the deck

Structural source, of which emission is based on ASJ-RTN Model

**fills: position, emission, and the propagation of the structural source  
— every cell is now implemented**

**3 stacked PRs · 285 existing tests and 211 additional tests across 5 modules, all passing**

Guarded branches: with no bridge input the results are byte-identical to today, and no existing test changed

# What's still open

- **Known limitation: multi-deck stacks**

- a source under 3+ stacked decks needs the splice to recurse over several lower edges
- today only a single edge — documented, not yet built



Multi-level viaduct: several lower edges at once  
photo: pexels-iban-lopez-luna

- **Open question: the core has no seam for a new obstacle**

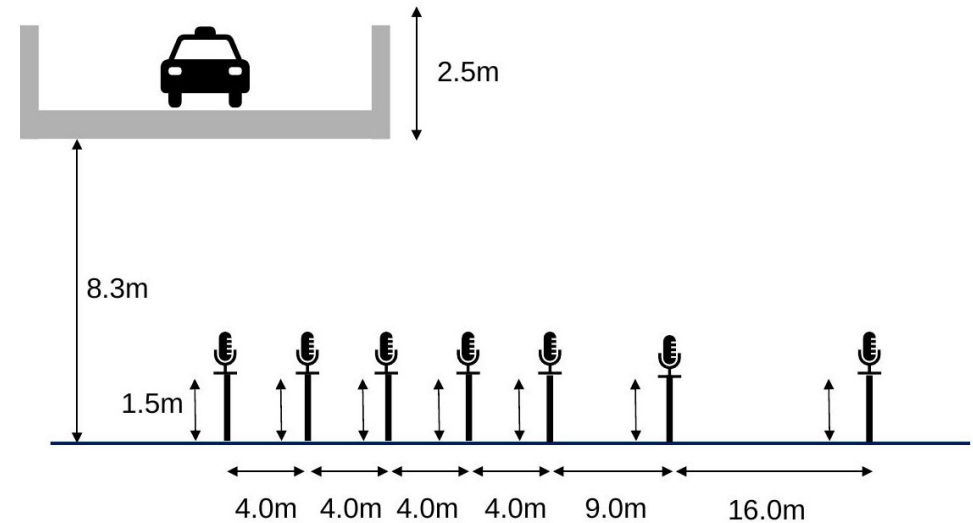
- bridges needed conditional branches

- **Refactoring toward maintainable code**

- divide the responsibilities — one class, one job: the ray-tracing loop split into ~9 collaborators (`PathFinder` 1142 → 254 lines)
- narrow the scope each class exposes — `ProfileBuilder`'s wide public API becomes separate building / wall / ground / topography services
- grow the developer documentation with it — documenting the `PathFinder` was how the bridge design was understood

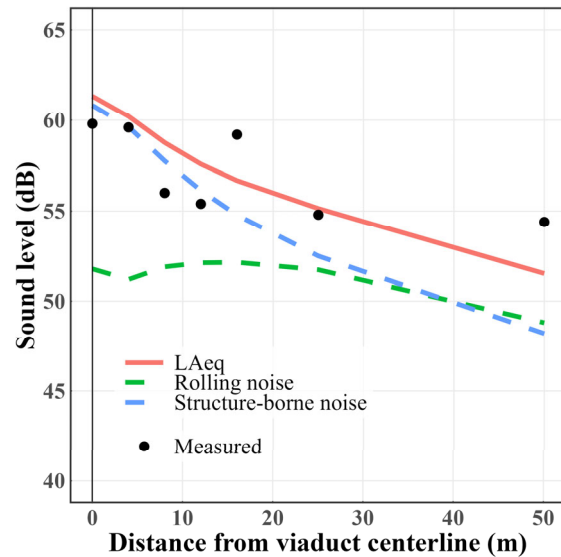
# Measured vs. predicted: Conditions

- **Site: Warashina River Bridge, National Route 1 (Shizuoka)**
  - Structure: steel deck plate-girder bridge
  - Traffic volume: 1,296 light / 330 heavy vehicles
  - Travel speed: 60.5 km/h
- **Measurement conditions**
  - 1/3-octave band sound levels
  - 10-minute duration, sampled 10 times per second
  - Noise level and traffic volume recorded simultaneously
- **Prediction conditions**
  - Prediction run with the integrated model (NoiseModelling extended with the ASJ viaduct calculation)
  - Traffic volume measured during the survey used as input
  - Bridge geometry and barrier height from 3D point clouds
    - *Estimated: barrier 2.5 m; viaduct 8.3 m high, 13.2 m wide*



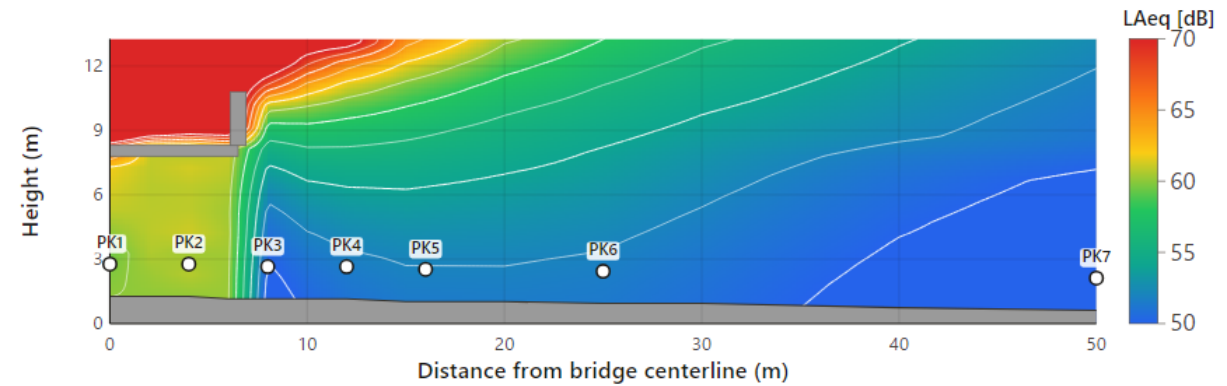
Microphones were arrayed perpendicular to the bridge axis.

# Measured vs. predicted: Results

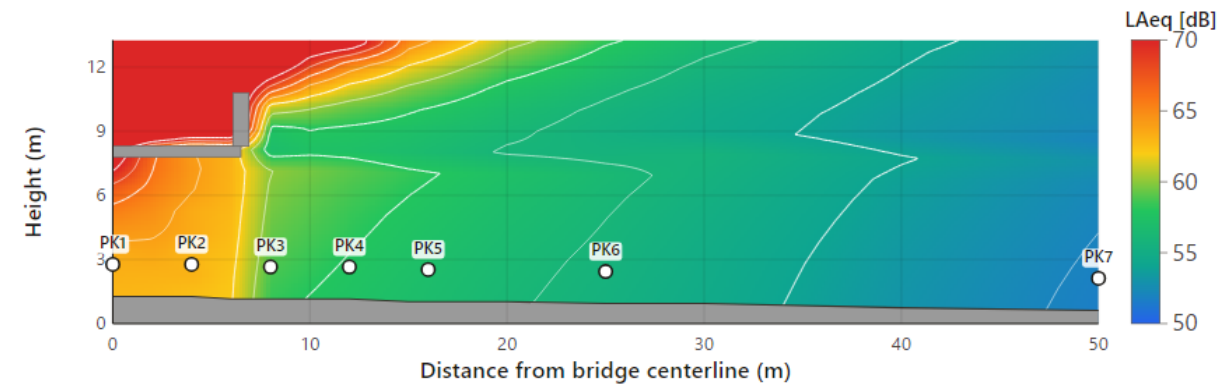


- **Predicted levels follow the measured trend with distance from the bridge**
  - Predicted LAeq combines the CNOSSOS-EU traffic contribution (ROAD) and the ASJ structural contribution (BRIDGE)
  - CNOSSOS-EU alone was insufficient; adding structural noise greatly improved accuracy
- **Enables mapping the level distribution on a vertical cross-section around the bridge**

Without structural noise



With structural noise



# Conclusion

- **Bridges are now modelled end to end in NoiseModelling**
  - deck geometry, on-deck traffic, reflection and double diffraction under the deck, and ASJ structural noise
  - three stacked, guarded PRs on current main: with no bridge input, existing scenes are untouched
- **Better predictions, and a new kind of output**
  - markedly closer to field measurements than CNOSSOS-EU alone
  - vertical-section level distributions around a bridge can now be drawn
- **Proven slowly, ported quickly — and open for discussion**
  - about a year of development on the 5.x fork, about a week to re-implement on current main with an AI coding agent
  - open items: multi-deck stacks, and whether the scripts should grow a seam for new obstacle types

Fork: [github.com/jtagusari/NoiseModelling](https://github.com/jtagusari/NoiseModelling) (fix) · PRs: feat/bridge → feat/bridge-v1 → feat/bridge-v2  
Contact: [tagusari.jun.mc@yokohama-cu.ac.jp](mailto:tagusari.jun.mc@yokohama-cu.ac.jp)